

the central mystery



mass and energy

The distinction between waves and particles in the everyday world is highlighted by a surfer ("particle") in a definite location riding on a spread-out wave.

background image

Because of the way waves spread out, two light waves can interfere with each other to make striped patterns.

One of the greatest physicists of the 1900s, Richard Feynman, described the double-slit experiment, which proved among other things that light travels as a wave, as "the central mystery" of quantum physics. There is no doubt that the experiment is correct and that

it can be used to calculate the wavelength of light. Light is a wave. But in the early decades of the 1900s, repeated experiments, such as studies of the photoelectric effect, showed just as surely that light is made up of particles—photons. Many of these experiments were carried out by Robert Andrews Millikan, who opposed Einstein's interpretation of the photoelectric effect and set out to prove Einstein wrong. Ironically, his excellent work conclusively demonstrated that Einstein was correct. The evidence was much more compelling than it would have been had Millikan been trying to confirm the theory.

a probability model

So how can the results of the double-slit experiment be explained in particle terms? The short answer is: they can't. Nothing in our everyday experience equips us to understand what is going on. The bottom line is that the way that events occur in the quantum world is utterly different from the everyday world of our experience, and that the truth can only be fully expressed in the



Albert Einstein

(1879–1955) is best known for his two theories of relativity, but his contributions

to quantum theory were also profound, and he received the Nobel Prize for his work on the photoelectric effect. After graduating in 1900, he completed his Special Theory of Relativity and his work on the photoelectric effect while working as a patent officer, and he did not obtain a university post until 1909.

diffraction pattern you see when waves pass through two holes. But if the "rocks" are photons, and the holes are very small, you do get a diffraction pattern. It is as if there is a wave associated with each photon. The waves from all the photons interact and make a diffraction pattern, which can be thought of as a probability pattern in that photons will most probably end up where the waves have interfered to produce peaks in the diffraction pattern, and less likely to end up where the interference has produced a canceling of the waves.

However, there is no way to predict in advance where any individual photon will end up. This is strange enough when we think in terms of photons. But it gets even stranger when we think in terms of electrons.

mathematical equations that enable physicists to predict things like the energy levels of electrons in an atom or the brightness of the stripes in a diffraction pattern.

One model that many people have found useful as an aid to understand these ideas involves waves of probability. If you threw rocks at a wall with two holes in it, the result would be two piles of rocks, one behind each hole. You would not get anything like the

strange behavior

If two people threw a lot of rocks onto a heap, you would be amazed if they lined up in stripes like the interference patterns made by waves.

